

Final Project: Classifying the American Sign Language Alphabet

✔ **Video:** Final Project: Classifying the American Sign Language Alphabet
2 min

✔ **Reading:** Project Introduction: Introducing the ASL Dataset
5 min

📖 **Practice Quiz:** Project Part 1 - Investigate and Prepare Your Data
30 min

📖 **Quiz:** Project Part 2 - Train and Evaluate a Model
1h 20m

📖 **Practice Quiz:** Project Part 3 - Classify New, Unlabeled Images
20 min

Summary and Next Steps

Project Introduction: Introducing the ASL Dataset

This dataset contains images of signed letters in the American Sign Language (ASL) alphabet. This dataset consists of 24 classes, one for each letter excluding "J" and "Z". The images portray different speakers with different fluency levels and contain various backgrounds. Some images show the speaker's left hand.

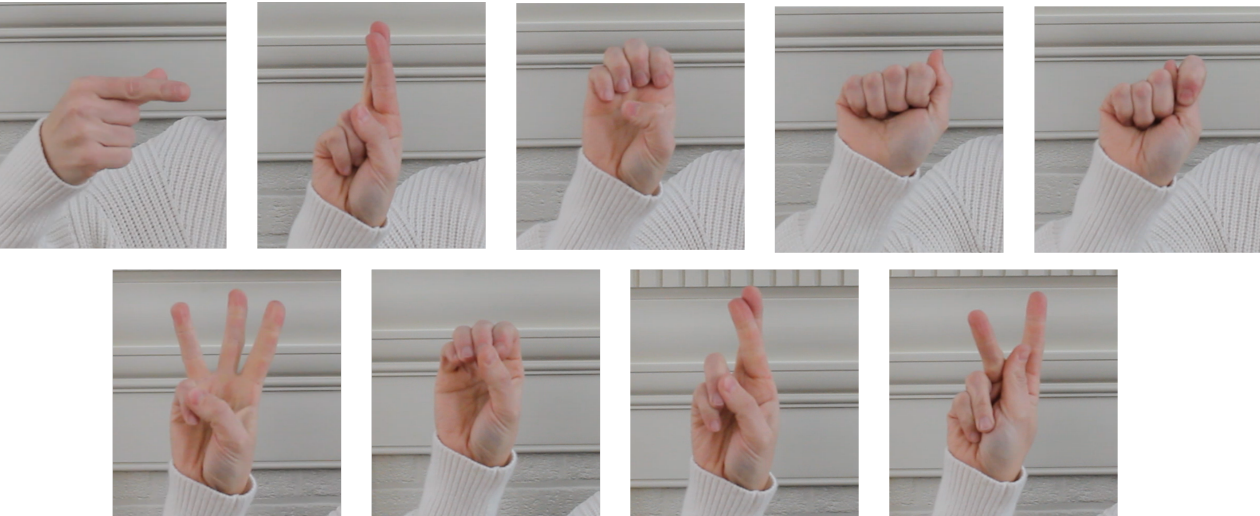


The goal of this project is to create a classifier that can classify the images in the test set with an accuracy of at least 90%.

The images have already been labeled and split into training and test sets in the "Data/MathWorks Created/ASL Alphabet/Classification" folder of the course files.

Your tasks for this project are the following:

- **Part 1:** Investigate and Prepare Your Data
 - Recall that the best practice is to include a randomized validation set. For this project, specify **80% of the images in the "Train" folder for training and the remaining 20% for validation.**
 - You will need to resize the images to perform transfer learning.
- **Part 2:** Train and Evaluate a Classifier
 - You must train a model that classifies the **test data** with at **least 90% accuracy.**
 - You may want to visualize your results to understand better how your model is performing.
 - It may take several attempts with different training options to achieve the required test accuracy.
- **Part 3:** Classify New, Unlabeled Images
 - Once you have a trained classifier, your final task is to apply it to new, unlabeled images from one of the speakers. The unlabeled images are in the "Data/MathWorks Created/ASL Alphabet/Classification/Unlabeled" folder of the course files.



After you have completed each part, complete the associated assessment before proceeding to the next step.

Go to next item ✓ Completed

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